

Zhengyang Yu

Curriculum Vitae

San Diego & Bay Area

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA

Expected Graduation: June 2028; GPA: 4.0/4.0

Relevant Coursework: Advanced Data Structures, Design and Analysis of Algorithms, Systems Programming, Computer Organization, Machine Learning

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*

- Owned the end-to-end design and development of a C++ product on ESP32-S3 that authenticates multiple accounts, retrieves live data over HTTPS, and presents it through an animated touch-controlled interface with configurable alerts.
- Implemented concurrent FreeRTOS workers, adaptive and rate-limit-aware polling, and PSRAM-backed rendering; added structured logging, runtime diagnostics, and iterative testing to improve performance and reliability.

2025 **HTML to ePub Converter**, *Personal Project*

- Engineered a Python content-processing pipeline with modular HTML parsing, metadata embedding, and structured error handling for reliable ePub generation.

2024 **Home Assistant Presence Plugin**, *Personal Project*

- Built a Python integration that converts entry and exit sensor events into real-time room-occupancy state.
- Designed an event-driven REST API workflow with explicit state transitions for reliable smart-lighting automation and easier testing and debugging.

Skills

Programming C++, Python, Java, C

Development Requirements analysis, API integration, interactive user interfaces, software testing, technical documentation

Engineering Data Structures, Algorithms, Object-Oriented Design, concurrent and event-driven systems, debugging, performance optimization

Tools Git, Docker, Linux, FreeRTOS, PlatformIO, VS Code, HTTP/HTTPS, REST APIs, structured logging

AI Tooling Codex-assisted implementation, prompt iteration, refactoring, debugging, code validation, and documentation

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch

Diagnosed computer and software issues and translated technical solutions into clear guidance for community members.